

Renewable Energy Storage Readiness Framework (RESRF)

Project Summary & Methodology

A Decision-Support Framework for Renewable Energy Integration and Battery Energy Storage Planning Across Indian States

1. Executive Summary

India's transition towards renewable energy has accelerated significantly over the past decade, driven by ambitious national targets, policy reforms, and sustained investments in solar, wind, hydro, and bioenergy. This expansion has positioned the country among the world's leading renewable energy markets. However, the pace of development has remained uneven across states. While a few resource-rich states, including Gujarat, Rajasthan, Tamil Nadu, Karnataka, and Maharashtra, have emerged as major centres of renewable energy deployment, several other regions continue to operate well below their technical potential. As renewable energy penetration increases, ensuring grid reliability and planning future investments in energy storage have become equally important challenges alongside expanding generation capacity.

Although India has developed a strong ecosystem of publicly available energy datasets, these resources remain fragmented across multiple agencies. Information on installed renewable capacity, electricity generation, solar and wind potential, electricity demand, demographic characteristics, and emerging storage policies is published independently by organisations such as the Ministry of New and Renewable Energy (MNRE), the Central Electricity Authority (CEA), the National Institute of Solar Energy (NISE), the National Institute of Wind Energy (NIWE), the Census of India, and various State Electricity Regulatory Commissions. These datasets differ in reporting periods, formats, geographical classifications, and naming conventions, making integrated state-level analysis both time-consuming and difficult.

As a result, answering practical planning questions often requires significant manual effort. For instance, identifying states that possess high renewable resource potential but relatively low deployment, or recognising regions where growing electricity demand may justify future Battery Energy Storage System (BESS) investments, cannot be achieved using any single official dataset. Instead, analysts must consolidate information from multiple sources before meaningful comparisons can be made. This fragmentation limits the ability of policymakers, researchers, and investors to develop evidence-based strategies using a consistent analytical framework.

The Renewable Energy Storage Readiness Framework (RESRF) is proposed as an attempt to address this gap. Rather than analysing renewable energy indicators independently, the framework aims to integrate multiple state-level datasets into a unified analytical model that supports a more comprehensive assessment of renewable energy deployment, generation performance, resource availability, electricity demand, demographic context, and future storage requirements. By bringing these indicators together within a single validated dataset, the project seeks to establish a consistent analytical foundation for comparing states and identifying opportunities for future renewable energy and storage investments.

The project will begin by developing a structured master dataset using information collected from official government publications and verified public sources. Following data validation, standardisation,

and feature engineering, the integrated dataset will be used to develop an interactive Power BI dashboard designed to communicate the current renewable energy landscape across Indian states. The initial dashboard focuses on descriptive analytics by presenting key performance indicators, spatial distribution, installed capacity, renewable generation, and regional variations. Subsequent dashboards will progressively extend this analysis towards evaluating renewable energy readiness, identifying investment opportunities, and supporting future battery energy storage planning.

This document presents the motivation behind the project, outlines its objectives, describes the proposed analytical methodology, and explains how the different datasets will be integrated to build the Renewable Energy Storage Readiness Framework. It also provides an overview of the planned dashboard design and the analytical narrative that will guide the project's future development.

3. Objectives

The Renewable Energy Storage Readiness Framework (RESRF) has been conceptualised to develop an integrated analytical framework for understanding the renewable energy landscape across Indian states. Rather than examining individual indicators in isolation, the project seeks to establish a comprehensive evidence base that supports more informed planning, investment, and policy decisions. The specific objectives of the project are as follows:

Develop an integrated state-level renewable energy database by consolidating information from multiple official sources, including renewable energy capacity, electricity generation, resource potential, electricity demand, demographic characteristics, and policy-related datasets into a single validated analytical framework.

Ensure the quality, consistency, and traceability of the integrated dataset through systematic data validation, standardisation, and documentation, enabling every analytical indicator to be linked to an official and verifiable source.

Develop meaningful analytical indicators that enable fair comparison across states with differing geographical, demographic, and resource characteristics. These indicators are intended to move beyond absolute values and provide deeper insights into renewable energy deployment, resource utilisation, and overall performance.

Design an interactive decision-support dashboard that enables policymakers, researchers, and investors to explore renewable energy patterns at multiple levels of detail, supporting data-driven assessment of regional opportunities, disparities, and emerging storage requirements.

Establish a scalable and reproducible analytical workflow that can be updated periodically as new datasets are released by official agencies, ensuring that the framework evolves from a one-time analytical exercise into a reusable decision-support platform for future renewable energy planning.

4. Dataset Overview

The master dataset (RESRF_Master_Dataset.xlsx) covers all 28 States and 8 Union Territories on 22 state-level variables spanning six analytical domains, assembled from eleven distinct primary sources. The table below summarises coverage; full provenance for every column is recorded in the workbook's own Data_Dictionary, Merge_Log and External_Data_Log sheets.

Domain	Variables	Primary Source
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Installed Capacity	Solar, Wind, Small Hydro, Large Hydro, Bio-power, Total Renewable Capacity (MW)	MNRE — State-wise RE Installed Capacity, 30.06.2026
Generation	Solar, Wind and Total Renewable Generation, June 2026 (MU)	CEA — RE Monthly Generation Report, June 2026
Demand	Energy Requirement, Energy Availability, Energy Deficit, June 2024 (MU)	CEA — Power Supply Position report
Resource Potential	Solar Potential (GWp), Wind Potential at 120m hub height (MW)	NISE / NIWE potential assessments
Demographics	Population, 2011 Census	Census of India 2011
Policy & Storage	State resource-adequacy regulations, national ESS policy timeline, national BESS project pipeline, national peak demand	SERC orders, MNRE/MoP policy notifications, MNRE BESS status reports, CEA

Two limitations of the underlying source data are carried through transparently rather than papered over. First, the CEA power-supply-position series used for state-wise Energy Requirement and Availability is current only to June 2024, roughly two years behind the capacity and generation figures; every derived metric that uses it (Energy Deficit, Renewable Intensity) is flagged with this period caveat in the Data_Dictionary. Second, Jammu & Kashmir's Census population and demand figures are pre-bifurcation and therefore include Ladakh, which is left blank on those specific variables rather than split by assumption — Ladakh's installed capacity, by contrast, is reported separately in the MNRE source and is fully populated. Neither gap is filled by estimation; both are documented as open items for the next data-refresh cycle.

5. Methodology

The Renewable Energy Storage Readiness Framework (RESRF) is being developed through a structured analytical workflow that transforms data from multiple official sources into an integrated decision-support framework. The methodology is designed not only to consolidate diverse datasets but also to ensure that the resulting analysis is transparent, reproducible, and suitable for future updates as new renewable energy statistics become available.

The project begins with the collection of renewable energy data published by various government agencies, including the Ministry of New and Renewable Energy (MNRE), the Central Electricity Authority (CEA), the National Institute of Solar Energy (NISE), the National Institute of Wind Energy (NIWE), the Census of India, and other relevant public sources. Since these datasets are published independently and differ in structure, reporting formats, and update cycles, they first require careful organisation before any meaningful analysis can be undertaken.

The next stage focuses on preparing a unified analytical dataset. Each source is reviewed independently to identify inconsistencies, remove duplicate or aggregated records, standardise variable names, and ensure that state and union territory names remain consistent across all datasets. This process is intended to minimise errors during integration and establish a common structure that supports reliable state-level comparison.

Once the datasets have been standardised, they are combined into a single master dataset that serves as the foundation for the project. Particular emphasis is placed on validating every merged variable against its respective source to ensure that the analytical framework remains transparent and that each indicator can be traced back to an official publication. Rather than estimating missing information, the project relies exclusively on verified government data wherever available.

After establishing the master dataset, additional analytical indicators are derived to support more meaningful comparisons between states. Instead of relying solely on absolute values such as installed capacity or electricity generation, the framework introduces relative measures that account for population, renewable resource availability, and energy demand. These derived indicators are intended to provide a more balanced assessment of renewable energy performance and help identify regions with significant untapped potential or emerging storage requirements.

The validated dataset will then be used to develop an interactive Power BI dashboard that communicates the findings through a series of connected visualisations. The dashboard is designed to move beyond descriptive reporting by allowing users to explore renewable energy patterns across different geographical regions and analytical dimensions. Rather than presenting isolated charts, the visualisations are intended to collectively support a coherent narrative, beginning with the current renewable energy landscape and progressively expanding towards questions related to storage readiness and future investment priorities.

Throughout the project, particular importance is placed on maintaining data quality, consistency, and reproducibility. The analytical workflow has therefore been designed so that future updates released by official agencies can be incorporated with minimal restructuring, allowing the framework to evolve as new renewable energy data becomes available.

7. Dashboard Storytelling

The dashboard is designed to communicate a coherent analytical story rather than present a collection of independent visualisations. It is intended to guide users from a high-level understanding of India's renewable energy landscape towards more detailed state-level insights that can support planning and decision-making. The overall design follows a layered approach, allowing users to move from national trends to regional comparisons and finally to indicators that may inform future renewable energy and battery storage planning.

7.1 Renewable Energy Capacity

The first component of the dashboard focuses on the geographical distribution of installed renewable energy capacity across Indian states and Union Territories. It aims to identify where renewable energy deployment has been concentrated and to highlight regional variations in capacity development. Rather than ranking states solely by installed capacity, this view provides the foundation for understanding broader patterns of renewable energy growth and identifying regions that may require greater policy attention or future investment.

7.2 Renewable Energy Generation

Installed capacity alone does not necessarily reflect how effectively renewable resources are being utilised. The second component therefore complements capacity information with renewable electricity generation, enabling users to compare infrastructure development with actual energy production. This comparison is intended to provide a more balanced understanding of renewable energy performance across states and to identify regions where further investigation into generation efficiency or resource utilisation may be required.

7.3 Regional Comparison

The third component introduces comparative indicators that account for differences in population, renewable resource availability, and energy demand. By normalising key variables, the dashboard aims to support more meaningful comparisons between states with diverse geographical and demographic characteristics. This perspective helps distinguish differences arising from scale from those that reflect genuine disparities in renewable energy development and access.

7.4 Renewable Energy Storage Readiness

The final component represents the long-term objective of the framework. It is intended to bring together multiple analytical indicators—including renewable resource potential, electricity demand, renewable deployment, and other supporting variables—to identify states where future Battery Energy Storage Systems (BESS) may play a greater role in facilitating renewable energy integration. As comprehensive state-level BESS datasets are currently limited, the framework does not seek to predict storage deployment directly. Instead, it proposes a set of evidence-based indicators that can serve as an initial analytical framework for identifying regions where storage investments may become increasingly important as renewable energy capacity continues to expand.

Together, these components are intended to transform the dashboard from a descriptive reporting tool into an analytical decision-support platform. While the first dashboard focuses on establishing the current renewable energy landscape across India, subsequent dashboards will progressively build upon this foundation by incorporating additional indicators, deeper analytical comparisons, and more targeted insights related to renewable energy planning and storage readiness.

8. Expected Outcomes

The proposed Renewable Energy Storage Readiness Framework (RESRF) is expected to establish a structured analytical foundation for understanding renewable energy development across Indian states. By integrating multiple official datasets into a single framework, the project aims to simplify state-level analysis and support more consistent comparisons of renewable energy capacity, generation, resource availability, electricity demand, and related indicators.

The integrated dataset and dashboard are expected to provide a clearer picture of regional variations in renewable energy development, enabling policymakers, researchers, and investors to identify states with significant renewable resource potential, emerging infrastructure requirements, and opportunities for future investment. Rather than relying on isolated indicators, the framework seeks to encourage a more holistic approach to evaluating renewable energy readiness.

As the project evolves, the framework is also expected to serve as a preliminary analytical tool for exploring battery energy storage opportunities. By combining multiple technical and contextual indicators, it aims to support the identification of states where further assessment of storage infrastructure may be warranted. The framework is intended to complement, rather than replace, detailed techno-economic studies and project-level feasibility assessments.

Another anticipated outcome is the development of a transparent and reproducible analytical workflow that can be updated as new data becomes available from official government sources. This will allow the framework to remain relevant over time while providing a consistent basis for future renewable energy assessments.

Overall, the project aims to demonstrate how integrated data analytics and interactive visualisation can strengthen evidence-based decision-making in India's renewable energy sector, providing a scalable foundation for future research, policy analysis, and infrastructure planning.

9. Future Scope

The current project establishes the foundation for an integrated renewable energy analytics framework. As additional datasets and more recent information become available, the framework can be expanded to provide deeper analytical insights and support a wider range of policy and investment decisions.

One important area for future development is the incorporation of regularly updated renewable energy and electricity demand data. Periodic data updates would enable the framework to move beyond a static assessment and support trend analysis, allowing users to monitor how renewable energy deployment, electricity demand, and storage requirements evolve over time. Another natural extension is the integration of state-level Battery Energy Storage System (BESS) information as official datasets become available. Incorporating project pipelines, installed storage capacity, and operational characteristics would strengthen the framework's ability to assess storage readiness using direct evidence rather than proxy indicators.

The analytical framework can also be enhanced through the development of a composite Renewable Energy Storage Readiness Index, combining multiple technical, infrastructural, and demand-related indicators into a single decision-support metric. Such an index would provide a consistent basis for comparing states and identifying regions that may require priority attention for renewable energy and storage investments.

As the project matures, the data processing workflow can be further automated to simplify the integration of newly published datasets from official government agencies. This would improve the efficiency of future updates while ensuring that the analytical framework remains transparent, reproducible, and current.

Finally, the framework has the potential to expand beyond descriptive analytics by incorporating forecasting techniques, scenario analysis, and predictive modelling. These additions would enable the project to evaluate future renewable energy growth, changing electricity demand patterns, and potential storage requirements under different development pathways, making the framework increasingly valuable as a long-term decision-support tool.

10. References

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